

APPROVED

MSc Drilling and Well Engineering
School of Computing, Engineering and Technology

Version	4.0
Course Type	Postgraduate
Award Type	Master of Science - MSc
Awards	
No Course Award Assigned	
Code	0323
SCQF Level	Level 11
QAA Subject Benchmark	Engineering (2023)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2026 September as of (September 2026)

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100184	general or integrated engineering	general or integrated engineering	100%
		Total	100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The course has been developed in close collaboration with industry to enable students to: > develop a critical approach to Drilling and Well Engineering > gain a broad and sound understanding of scientific, engineering and management aspects relating to selected key areas of Drilling and Well Engineering > develop and utilise skills in integrating knowledge drawn from different subject areas, through coursework, case studies, research and project work > encourage innovative thinking, which can be applied to the discipline of Drilling and Well Engineering > develop initiative, creativity, inter-personal skills and a high level of independent critical judgment. > develop a critical approach to operate a sustainable enterprise business so as to be viable, grow and earn a profit.
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What the Course Involves	Details of the modules contained in each semester can be seen further in this document. Regarding the course structure: Full Time Students - For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (project, taken in May) - For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 3 (project, taken in January) Part Time Students - For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (taken in September), Semester 4 (taken in January), Semester 5 (project, taken in May and spanning 2 semesters) - For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 4 (taken in January), Semester 3 (taken in September), Semester 5 (project, taken in January and spanning 2 semesters) For all modes of delivery and key to the part-time online learning delivery, extensive use is made of RGU's Virtual Learning Environment (Moodle) which allows interaction with teaching staff and other students. Detailed lecture notes with activity questions will be through the Moodle through an on-line tutorial forum. Should sufficient demand arise, a module may be delivered as an in-class, face-to-face lecture and tutorial based course, with case study work, plus private study and discussion.
Course Collaborations	
No collaborations listed	
Introduction to the Course	This course provides a sound understanding of the scientific, engineering and management aspects that relate to the disciplines of drilling and well engineering. This course has been designed to educate the next generation of drilling and well engineers, and has evolved in response to the developing needs of the oil and gas industry, looking at how they can implement new mechanism to their end-to-end value chains for a sustainable future. It focuses on the concept of effective well construction, and aims to develop high levels of professional capability in the key areas of well design, drilling and operations management, as well as, sustainability strategy capability guides to build innovative business case scenarios and develop sustainability roadmaps. This course provides the knowledge required for industry, including sustainability and innovation (developing new thinking/technology in oil and gas industry to achieve low carbon and NetZero).The course will provide students with an understanding of the market drivers for sustainability and the application of engineering relevant to upstream oil and gas engineering and support them in developing innovative ideas to address the United Nations Sustainability Goals (UNSDGs). The MSc Drilling and Well Engineering course is accredited by the Energy Institute to ensure that students meet the competencies established by the Engineering Council.
	The MSc Drilling and Well Engineering is focused on the planning, construction and delivery of wells designed to facilitate the extraction of hydrocarbons. The course is designed to give an initial broad coverage of the industry, followed by specialisation in the specific discipline. The course aims to: - Develop the attributes expected of Drilling and Well Engineering Masters professionals. - Deliver proof of ability to evaluate and synthesize at the Society of Petroleum Engineers (SPE) required knowledge depth for all specified skills, as specified in the SPE matrix for Petroleum Engineers. - Critically evaluate the considerations for delivering a United Nation Sustainable Development Goal (UNSDG) and design an innovative business

Educational Aims of the Course	Postgraduate Certificate • To provide a broad-based and sound understanding of the scientific, engineering and management subjects relevant to selected areas of effective drilling and well engineering. Postgraduate Diploma • To develop a critical approach to the implementation of drilling and well engineering. • To provide a broad-based and sound understanding of the scientific, engineering and management subjects relevant to effective drilling and well engineering. • To encourage innovative thinking, which is then applied to the discipline of drilling and well engineering. MSc • In addition to the aims of the Postgraduate Diploma, to develop initiative, creativity, interpersonal skills and a high level of critical judgement in the application of appropriate and relevant principles during the Masters level project stage.
Awarding Institution	Robert Gordon University
Course Duration	Full-Time On-Campus Students Full-time 12 months with a September intake Full-time 16 months with a January intake Part-Time Online Students Part-time 28 months with a September intake Part-time 32 months with a January intake Students can choose to undertake a short-term (3 month) placement or a long-term (12 month) placement. All placements require approval from the School prior to commencement. If a student goes on placement, the duration of the course will be extended by the duration of the placement.
Distinctive Features of the Course	The course combines academic engineering study with current, real-world industry practice. This is achieved through the knowledge and skills of the School of Computing, Engineering, and Technology full-time staff. Students on the course work on real field data where appropriate and feasible, as well as having the use of the drilling simulator during the course of their studies. External Guest Lecturers are invited to give talks, industrial visits are arranged on this course. The course is accredited by the Energy Institute.
	Students may elect to undertake a short placement or a year-long placement. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement.

Placement/Professional Experience	While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short placement. This normally comprises 4 - 16 weeks of continuous, full-time supervised work experience. The assessment for a short placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short placement will normally receive at least 1 video call with their Academic Tutor. As an alternative, students may undertake a year-long placement. The year-long industrial placement normally comprises of 30 - 45 weeks of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. Placements must be undertaken after the completion of all taught modules and begin before the start of the final project. A student may choose to either undertake a placement at the same time as their final project or delay the start of their final project until the placement has finished. The course leader must approve this decision. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 180 credits achievable from the other modules. Successful completion is not required for progression or graduation. The credits achieved for placement may not be used to compensate for any credits from the other modules.
Opportunities for Further Study	Graduates are qualified to apply for PhD studies in the School of Computing, Engineering and Technology.
External and Internal Indicators of Quality and Standards	Internal references include: • University Mission Statement External references include: • QAA Subject Benchmark for Engineering • SCQF Qualification Descriptors, Level 11

	• Energy Institute Accreditation • Peer review of staff research High quality Drilling and Well Engineering graduates are highly sought after.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Applicants normally hold a 2:2 undergraduate honours degree or its equivalent in Engineering or a relevant discipline such as Geology, Physics, Chemistry or Mathematics. Applicants whose first language is not English should have an IELTS of 6.5 overall, with no component lower than 5.5, or an equivalent.
Alternative Qualifications for Entry	Applicants with a lower degree classification and / or relevant work experience will be considered on a case by case basis.
Interview Procedures	Applicants are not usually interviewed although some may be if they do not meet the standard entry requirements.
Course-Specific Regulations	The following module has been identified as being central to the objectives of the course and, therefore, failure can not be condoned: CEM100 MSc Project Students are awarded a Postgraduate Certificate Oil and Gas Engineering on exit from the programme with 60 credits at SCQF 11 level from Stage 1. Students are awarded a Postgraduate Certificate Drilling and Well Engineering on successful completion of 4 modules (60 SCQF credits) from Stage 1, at least 30 of which must be from Stage 1, Semester 2 modules. All students, regardless of mode and start date, will typically undertake resits in the summer exam period. Exceptionally students who start in January may be offered a resit opportunity after completing two semesters, in April of the following year.

Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: • To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.

Course-Specific Learning Experience	The origin of the course lies with a change in the way the major oil and gas operators ran their operations. Following the various alliances and partnering initiatives, the major operators retracted to their core businesses. This meant that it was necessary to extend the responsibilities of their partners (the drilling and lead contractors) to include 'drilling engineering' and overall responsibility for the concept of 'well engineering'. This change in responsibility has led to the drilling contractors having difficulty in recruiting competent, professionally qualified drilling engineers. Traditionally, these engineers would have been recruited and trained by the major operators. They would have undergone a structured training programme including examinations. Subsequently, they would become responsible for tasks such as well design, and all aspects of drilling and workover operations. The current partnering and alliance arrangements have led to the lead/drilling contractor assuming the responsibility for these tasks. Drilling engineers employed by the oil companies are now often office based and responsible for such tasks as planning, budgeting, cost (reduction) engineering, conceptual planning and new technology implementation. This MSc course is designed to provide the required knowledge and underpinning theory for both the drilling contractor and major operator. The stimulus for this specialist course was an initiative from the drilling industry. Subsequent discussion with major operators and drilling contractors laid the foundations of the course. The course has been devised to be a sustainable, authentic, competency based MSc course which delivers the attributes expected of drilling and well engineering Masters professionals. It provides an equal standard of learning experience to all modes of delivery. It is based on the industry recognised Society of Petroleum Engineers (SPE) competency matrix for Petroleum Engineers. It delivers proof of ability to evaluate and synthesise at SPE minimum competence depth for all specified skills. The modules are taught by a mixture of staff from the University - to ensure that the material is current and relevant. All student materials, for full time and online learning part time students are available through Moodle the University's Virtual Learning Environment (VLE). Online learning part-time students make extensive use of the Moodle as a means of interacting with staff.
	The teaching and learning strategy for Drilling and Well Engineering Course aligns with that of the School of Computing, Engineering and Technology teaching and learning strategy which aims to produce industrially relevant professional engineers. In line with the university's teaching, learning, and the student experience strategy, this will enable students to raise their aspirations and thrive in their professional careers as global citizens. RGU continues to lead the way in addressing global sustainability challenges and recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas. Consequently, sustainability is integrated into all courses at the School of Computing, Engineering and Technology, equipping students with skills and competencies that are essential for their future careers. The teaching and learning strategy has been designed with reference to engineering council standards to ensure students develop the required professional competencies as they progress through the course. This includes a focus on sustainability with students tackling real-world engineering challenges whilst considering sustainable practice. The curriculum has been designed so that students develop their knowledge and understanding, practical skills, general cognitive skills, and key transferable skills at five levels corresponding to level 11 of the Scottish Credit and Qualifications framework. Projects and teamwork are embedded in the course. This helps students to integrate theory and practical skills. It also encourages creativity and the development of transferable skills, including the ability to work in a team and an understanding of sustainable practice, enhancing students' employability. Project work, both in a team and individually, helps to develop students as independent, enthusiastic learners. Whilst developing and delivering the curriculum the School of Computing, Engineering and Technology has strong interaction with industry. This relevant and engaging curriculum enhances the teaching and learning experience, helping to ensure that content remains relevant to today's engineering challenges and improves students' employability.

Teaching and Learning Strategy

The School of Computing, Engineering and Technology recognises the importance of providing a high-quality learning environment with appropriate use of technology in teaching and learning. It encourages and supports staff in the development of innovative and effective teaching methods and excellent academic support. Student feedback is valued and used to develop and enhance the teaching and learning experience. Students develop the competencies required by a graduate engineer through a wide variety of teaching and assessment methods. These are aligned with the learning outcomes of the modules and the professional competencies students are expected to develop. These are described in more detail in the following sections.

Knowledge and Understanding

Knowledge and understanding are developed through a number of methods, which are module and delivery mode specific, such as:

- Lectures and solving tutorial problems - for the development of knowledge and understanding of a subject.
- Connect and Reflect Week Activity, field trip and/or Industrial Visit.
- Seminars, workshops and guest lectures.
- Participating in online discussion forums.

Those direct teaching methods are usually supported by coursework and project work, and are supplemented with direct study by means of:

- Learning materials on Virtual Learning Environment (Moodle).
- Textbooks and other recommended learning materials.
- Journal articles and conference papers.

Practice: Applied Knowledge and Understanding

Practical skills are developed through carrying out practical activities in an appropriate learning context. This is achieved by several methods, which are module and delivery mode specific, such as:

- Practical exercises.
- Workshops.
- Individual and group projects.
- Participation in field trips or industrial visits.

Generic Cognitive Skills

Generic cognitive skills such as critical analysis, synthesis, evaluation, and problem solving, are developed through active learning processes, which are module and delivery mode specific, such as:

- Group-learning activities - seminars, tutorials or case studies.
- eLearning teaching material.
- Computer-based simulation.

- Problem-solving exercises.
- Individual and group projects.
- Practical activities and engineering application work.

Generic cognitive Skills are also developed by means of project work at various stages of the course.

The MSc Project stage involves independent project work. This allows the student to demonstrate capability in a range of intellectual skills linked to specialist knowledge, understanding and practical skills

Communication, ICT and Numeracy Skills

Communication, ICT and numeracy are developed through naturally arising opportunities within the curriculum, which are module and delivery mode specific. For instance:

- Written communication skills are developed through contributions to online discussion forums, developing answers to tutorial problems, coursework, essays and project reports.
- Oral communication skills are developed through interactions in lectures, tutorials and team activities, group projects, interactions and presentations in seminars and the presentations associated with projects.
- ICT skills are developed through the use of a wide range of ICT in learning and completing assignments.
- Numeracy Skills are developed through the interpretation, use and evaluation of a wide range of numerical data, obtained through theoretical study, coursework, practical work and projects.

Skills may also be developed through extra-curricular activities including:

- Work experience.
- Engineering Society.
- Social and cultural activities.

Autonomy, Accountability and Working with others

Autonomy, accountability, communication and working with others (soft and hard) are skills that could be transferred to employment and other context freely. These skills are developed through naturally arising opportunities within the curriculum, which are module and delivery mode specific. For instance:

- Team working skills are developed through collaborative exercises and case studies.
- Group presentations
- Group project plan
- Reflective coursework MSc Project stage involves independent project work.
- Demonstrate capability in a range of intellectual skills linked to specialist knowledge, understanding and practical skills

	The assessment of the MSc Drilling and Well Engineering course is based on the course structure and includes a mix of project work, coursework, and closed-book examinations. The type of assessment used depends on the course stage, SCQF level, learning outcomes, and module content. Each module is assessed using a single assessment method. The course assessment strategy consists of four 15-credit modules in the Postgraduate Certificate stage, followed by four 15-credit modules in the Postgraduate Diploma stage (PgDip), and a 60-credit individual MSc project (CEM100). The PgDip stage assessments include a combination of written examinations, coursework, individual reflective work, group work, and presentations tailored to the module content. Regular assessment planning reviews are conducted to avoid over-assessment. Formative assessments are integrated into all modules to support learning, with activities like group work, presentations, project plan development, and case study reviews. Lectures provide a foundation for these exercises, aiding in knowledge retention and fostering a supportive peer network within each cohort. The project allows many elements of the course to be integrated. The project activity is assessed by a thesis. Students are provided with a detailed project handbook that describes the various stages in the project and the means by which it is assessed. Students undertaking the course in online learning part-time mode can arrange to sit their examinations off-site. The University and the School have well-developed procedures for selecting suitable venues and making arrangements for invigilation. Knowledge and Understanding The primary means of assessment of knowledge and understanding is by means of closed book examinations. However, it may also be assessed by means of: • Closed book examinations. • Computer based assessment. • Oral Presentations Some elements of knowledge and understanding are also assessed by means of the MSc project. Practice: Applied Knowledge and Understanding Practical skills are assessed by the most appropriate means, depending on the nature of the practical work. These skills are assessed by means of: • Practical exercises and reports (e.g. for case studies and design exercises). • Project work. Generic Cognitive Skills

Assessment

Generic cognitive skills such as critical analysis, synthesis, evaluation, and problem solving, are developed through active learning processes, which are module and delivery mode specific, such as:

- Group-learning activities - seminars, tutorials, or case studies.
- Computer-based simulation.
- Problem-solving exercises.
- Individual and group projects.
- Oral presentation.

The MSc project provides the opportunity for students to integrate material covered in modules in the earlier part of the course and to demonstrate their ability within analytical, design and research skill

Communication, ICT and Numeracy Skills

Communication, ICT and numeracy are developed through naturally arising opportunities within the curriculum, which are module and delivery mode specific. For instance:

- Written communication skills are developed through contributions to online discussion forums, developing answers to tutorial problems, coursework, and project reports.
- Oral communication skills are developed through interactions in lectures, tutorials and team activities, group projects, interactions and presentations in seminars and the presentations associated with projects.
- ICT skills are developed through the use of ICT applications in learning and completing assignments.
- Numeracy Skills are developed through the interpretation, use and evaluation of numerical data obtained through theoretical study, coursework, practical work, and projects.

By the end of the Master's course, students are expected to effectively communicate complex ideas in a well-structured manner, both orally and in writing, utilizing computer-based tools.

Autonomy, Accountability and Working with others

Autonomy, accountability and working with others are skills that could be transferred to employment and other context freely. These skills are developed through naturally arising opportunities within the curriculum, which are module and delivery mode specific. For instance:

- Team working.
- Case studies.
- Group projects.

Skills may also be developed through extra-curricular activities including:

- Engineering Society activities.
- Social and cultural activities.

University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	• RGU Assessment & Feedback Standard • RGU Blended Learning Standard • RGU Digital Learning Standard All students are offered the opportunity to take part in an Induction Programme which provides an introduction to the School, the staff and a chance to get to know the other students on the courses. This primarily takes place at the start of the course but there are also ongoing events in the first semester. Further information is provided on the School's webpages and via the VLE, Moodle facility that enables a wide range of teaching and learning resources to be made available to students and members of staff. CampusMoodle also allows online marking and quick feedback to students. Module Coordinators are available to support students with course and module related issues and concerns. Dedicated administrative support will be available to deal with any queries students may have. Students will be supported through the School's integration of the institutional Team Around a Student (TAAS) approach, ensuring that every student has a point of contact who can provide advice, triage queries, and connect them with the most appropriate support and resources where issues may be affecting their studies or wider student experience.

	The University has a Study Support Service. This provides support in the areas of academic writing, maths and statistics, IT support, English language and Study Skills. In addition to this, students have access to support on effective use of digital learning provided by the Digital Learning Team in Department of Academic Quality and Learning Enhancement (AQLE). All students have access to a Student Handbook; this is provided in electronic format on the School's webpages. Students undertaking individual projects are provided with a detailed Project Handbook covering all aspects of the project operation and assessment. Extensive written guidance is also available via the VLE. All students have an account on the University computer network and can access PCs in the school and across the University. Specialist software is provided on PCs in Engineering. The support staff also assist students with administrative queries.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	Annual Course Appraisals are reviewed and approved by the Course Management Team and School Academic Board which meet at least once per semester. The Course Management Team considers teaching and learning issues arising from feedback generated by the External Examiner Annual Reports, Staff and Student Liaison Committees, Student Experience Questionnaires and the National Student Survey. The Course Management Team and School Academic Board also review all proposed changes to the structure and content of courses. The School's Teaching, Learning & Assessment Committee regularly looks at metrics such as student experience and student achievement to ensure that the courses are continuing to deliver the required standard set by the University. It also supports the ongoing enhancement of approaches to teaching, learning and assessment. The quality and standard of the assessment are ensured through a robust moderation process in the School. All summative assessments are moderated by an academic staff and an external examiner. The marking of all assessments is completed in full compliance with the University's Assessment policy.

University Student Feedback Mechanisms

The University and RGU:Union work in partnership to create an environment which stimulates:

- the participation of students, by empowering them to proactively provide views and opinions;
- a supportive learning community where students and staff engage in meaningful dialogue; and
- the engagement of students in the design of solutions and enhancements.

This approach is underpinned by the *Student Partnership Agreement* which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: [Student Representation & Partnership | RGU](#) .

There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience.

Mechanisms through which students are engaged and supported include:

- participation in student evaluation questionnaires and where relevant the *National Student Survey (NSS)* ;
- staff/student liaison arrangements at course/programme level;
- support for Student Representatives;
- student representation on Institution-Led Subject Review Panels and Validations;
- student representation on key University committees.

Course-Specific Student Feedback Mechanisms

The Staff Student Liaison Committee meets at least once per semester. The membership of the Committee consists of the Course Leaders, course administrators and student representatives. Student representatives are asked to collect comments from their class and they are also encouraged to liaise with Course Leaders/Year Tutors when issues arise (rather than waiting for the next Staff Student Liaison Committee meeting). Minutes of the meetings are posted on Moodle for all students on the course to access.

Awards**Stage 1**

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Postgraduate Certificate - PgCert	Oil and Gas Engineering	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Certificate - PgCert	Drilling and Well Engineering	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Diploma - PgDip	Drilling and Well Engineering	Aberdeen, United Kingdom	Exit Award		120
Master of Science - MSc	Drilling and Well Engineering	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	To analyse complex problems and demonstrate the knowledge and understanding to prepare and deliver a plan to manage a robust solution.	SC1 SC3	SCQF 11
LLO2	To demonstrate comprehensive knowledge of drilling & well engineering to the solution of complex problems in the context of broader disciplines of science, mathematics and engineering within which the course is set.	SC1 SC2	SCQF 11
LLO3	To demonstrate mastery of the skills to identify and classify principles and ideas in contemporary information sources and situations to professional standards; to analyse rigorously, effectively, critically and creatively; to cope with complexity, gaps and/or contradictions.	SC3	SCQF 11
LLO4	To confidently integrate theory with professional/vocational practice; evaluating theories, processes, solutions and outcomes critically and effectively; using the evaluations of others critically, reflectively and constructively.	SC2 SC3	SCQF 11
LLO5	To identify, define and resolve a range of environmental and societal impacts associated with complex problems to include the whole life cycle, minimising adverse impacts.	SC1 SC2 SC3 SC5 SC4	
LLO6	To demonstrate professional competence in the evaluation of available data using numerical/mathematical skills associated with the specialist areas covered by the course. This is also demonstrated in any data analysis associated with the programme of work	SC2 SC3 SC4	SCQF 11
LLO7	To demonstrate communication and presentation skills by engaging effectively both independently and working within a group; debating in a confident, professional manner; producing detailed critiques and coherent project reports to professional standards.	SC2 SC3 SC5	
LLO8	To apply relevant data analysis and mathematical skills, to the solution of complex problems, demonstrating mastery as this is done.	SC4	SCQF 11
LLO9	To work cohesively within a group demonstrating leadership and team working skills to develop an innovative project.	SC1 SC2 SC3 SC5	
LLO10	To demonstrate research process skills including initiation and completion of a thorough investigation via evaluation of methodological approaches, application of knowledge and theoretical approaches; communication and presentation of outputs of the project to meet a combination of societal, user, business and customer needs as appropriate.	SC1 SC2 SC3 SC5 SC4	

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7

SC3	Generic cognitive skills	SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A: Non Placement Route B: Short Placement Route C: Long Placement
Part Time Online	Route A: Non Placement Route B: Short Placement Route C: Long Placement

Stage 1 / Semester 1**Full Time On Campus**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Part Time Online

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CBM204	Data Analytics for Business Decisions		Sandra Cannon (sc11706)	Yes	3.0	15	Level 11
Full Time On Campus	ENM202	Facilities		Ruissein Mahon (rm8781)	Yes	3.0	15	Level 11
Full Time On Campus	ENM200	Subsurface		Mohamed Amish (ensmam)	Yes	3.0	15	Level 11
Part Time Online	ENM200	Subsurface		Mohamed Amish (ensmam)	Yes	3.0	15	Level 11
Full Time On Campus	ENM201	Wells		Mohamed Amish (ensmam)	Yes	3.0	15	Level 11
Part Time	ENM201	Wells		Mohamed Amish	Yes	3.0	15	Level

Online				(ensmam)				11
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Stage 1 / Semester 2**Full Time On Campus**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Part Time Online

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Full Time On Campus	ENM212	Advanced Well Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11
Full Time On Campus	ENM211	Drilling Operations Management		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11
Part Time Online	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
Full Time On Campus	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
Full Time On Campus	ENM206	Well Completions and Subsea Systems		Ruissein Mahon (rm8781)	Yes	4.0	15	Level 11
Part Time Online	ENM206	Well Completions and Subsea Systems		Ruissein Mahon (rm8781)	Yes	4.0	15	Level 11

Stage 1 / Semester 3**Full Time On Campus**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Part Time Online

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								Module Level
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	
Part Time Online	CBM204	Data Analytics for Business Decisions		Sandra Cannon (sc11706)	Yes	3.0	15	Level 11
Part Time Online	ENM202	Facilities		Ruissein Mahon (rm8781)	Yes	3.0	15	Level 11
Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								Module Level
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	
Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Stage 1 / Semester 4**Part Time Online**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								Module Level
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	
Part Time Online	ENM212	Advanced Well Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11

Part Time Online	ENM211	Drilling Operations Management		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11
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Stage 1 / Semester 5**Part Time Online**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time Online	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time Online	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Part Time Online	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Full Time On Campus	CBM204	Data Analytics for Business Decisions		Sandra Cannon (sc11706)	Yes	3.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	ENM202	Facilities		Ruissein Mahon (rm8781)	Yes	3.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	ENM200	Subsurface		Mohamed Amish (ensmam)	Yes	3.0	15	Level 11
1	Semester 1	Core	Part Time Online	ENM200	Subsurface		Mohamed Amish (ensmam)	Yes	3.0	15	Level 11
	Semester		Full Time				Mohamed Amish				Level

1	1	Core	On Campus	ENM201	Wells		(ensmam)	Yes	3.0	15	11
1	Semester 1	Core	Part Time Online	ENM201	Wells		Mohamed Amish (ensmam)	Yes	3.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM212	Advanced Well Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM211	Drilling Operations Management		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11
1	Semester 2	Core	Part Time Online	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM206	Well Completions and Subsea Systems		Ruissein Mahon (rm8781)	Yes	4.0	15	Level 11
1	Semester 2	Core	Part Time Online	ENM206	Well Completions and Subsea Systems		Ruissein Mahon (rm8781)	Yes	4.0	15	Level 11
1	Semester 3	Core	Part Time Online	CBM204	Data Analytics for Business Decisions		Sandra Cannon (sc11706)	Yes	3.0	15	Level 11
1	Semester 3	Core	Part Time Online	ENM202	Facilities		Ruissein Mahon (rm8781)	Yes	3.0	15	Level 11
1	Semester 3	Core	Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
			Full								

1	Semester 3	Elective	Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11
1	Semester 4	Core	Part Time Online	ENM212	Advanced Well Engineering		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time Online	ENM211	Drilling Operations Management		Mohamed Amish (ensmam)	Yes	2.0	15	Level 11
1	Semester 5	Core	Part Time Online	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 5	Elective	Part Time Online	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 5	Elective	Part Time Online	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	
All Applicable Locations	Online	Part Time	

Sources of Further Information

No sources of further information recorded
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